

KonetiQ

Advisors:

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Team:

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OUR GOAL

Develop a platform to monitor and manage corporate micromobility fleets alongside personal vehicle owners.



CONTEXT



1

What is mariaBike?

2

How is data currently stored?

3

What could be improved?

PROBLEMS

1

**Ineffectiveness of
Existing Solutions**

2

**Raw data without a
structured back-office**

3

Lack of extensibility

GOALS

Ingest **real-time telemetry** with an event-driven architecture, initially from the mariaBike fleet.

1

Manage each **bike's lifecycle** (registration, maintenance, usage history) and **aggregated fleet metrics**.

2

Expose extensible APIs for **value-added services**, with concrete implementations.

3

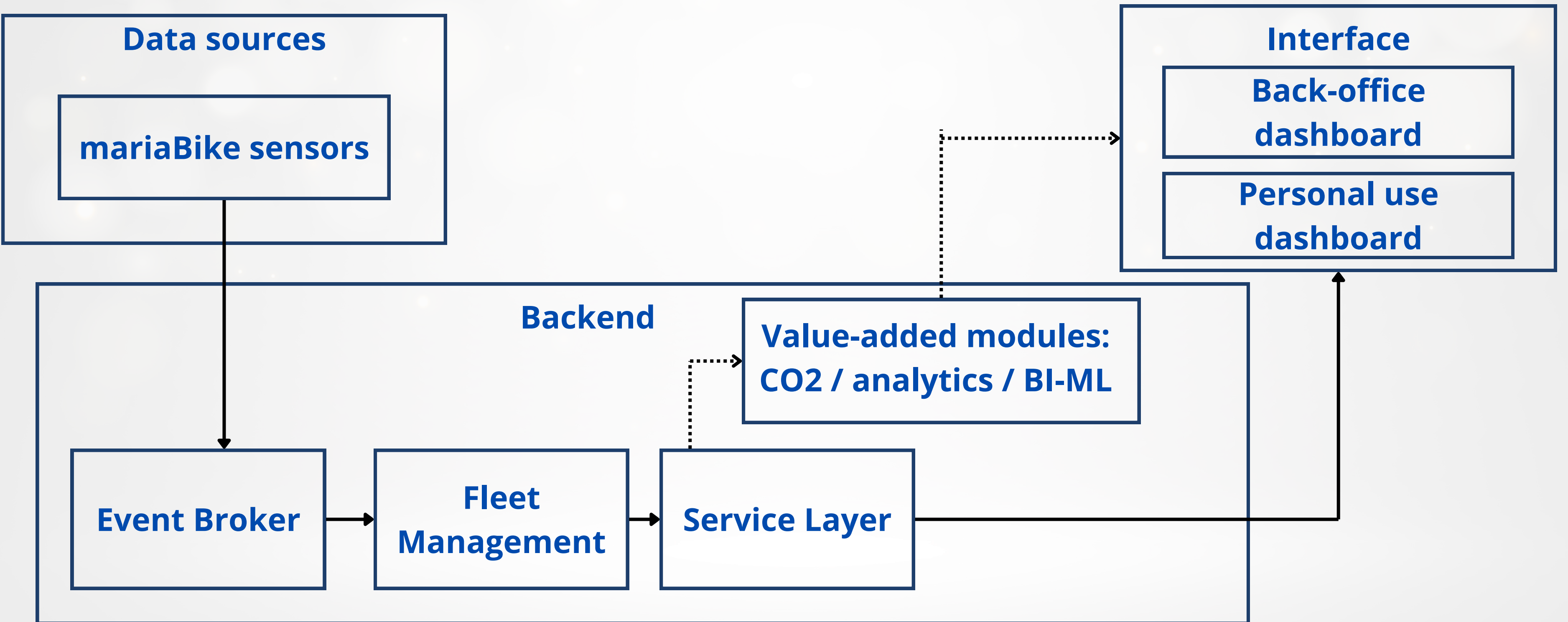
Deliver a back-office dashboard for operators to monitor **telemetry, lifecycle and service KPIs**.

4

Validate extensibility with one **BI reporting module** and one **Machine Learning module**.

5

HIGH LEVEL ARCHITECTURE



STATE OF THE ART

	Real-time Telemetry	Analytics / Business intelligence	ML-based module for anomaly detection	Vehicle lifecycle	Personal-use component	Open and extensible API	CO2 footprint tracking
Joyride	✓	✓	✗	✓	✗	✓	✗
Levy Fleets	✓	✓	✗	✓	✗	✓	✗
Comodule	✓	✓	✗	✗	✓	✓	✓
Trackap	✓	✓	✗	✓	✓	✗	✓
KonetiQ	✓	✓	✓	✓	✓	✓	✓

COMMUNICATION

1

Two meetings per week, one with the advisor

2

Whatsapp and Discord for everyday quick communication

3

Google Drive and Github for document sharing and version control

4

Jira for Scrum workflow tracking

CALENDAR

Milestone	Date	Tasks
M1	22/09-29/09	1 - Project Website & Visual Identity (João Pereira) 2 - GitHub Organization (Santiago) 3 - Jira Project (João Tomásio) 4 - Project Calendar & Planning (Maria) 5 - State-of-the-art & MariaBike Platform Audit (Everyone) 6 - Presentation Preparation (Guilherme)
M2	30/09-06/10	7 - Requirements Gathering & Scope Definition (João Tomásio - Jira logging and management; Maria - initial functional requirements structuring) 8 - Actors, Personas & Use Cases (João Pereira and Guilherme - operator and user profile definition; João Tomásio - core use case mapping) 9 - Event-Driven Architecture Initial Design (Santiago - core infrastructure; Maria - telemetry data streams; Johnny - UI integration touchpoint mapping)

CALENDAR

Milestone	Date	Tasks
	07/10-13/10	10 - Event-Driven Architecture Design & Kafka/MQTT Setup (Santiago - broker technical setup; Maria - data schemas and topic design) 11 - Non-Functional Requirements & Documentation (João Tomásio - constraints and security; João Pereira - usability requirements and UI/UX constraints) 12 - Presentation of Lifecycle Architecture (Guilherme - defense coordination and slide deck technical structuring with team-wide contributions)
M3	14/10-20/10	12 - Initial Mockups & Dashboard UI Concepts 13 - Main User Flows for Fleet Operators
	21/10-03/11	14 - Interactive Prototype & Design System 15 - Validation of Telemetry Data Ingestion

CALENDAR

Milestone	Date	Tasks
M4	04/11-24/11	Core Backbone & MVP: • Real-time Telemetry Ingestion Pipeline • Event Store & Database Setup • Individual Bike Registration & Lifecycle State • Aggregated Fleet Metrics API
	25/11-15/12	MVP Completion & Integration: • Service Layer: CO2 Footprint Tracking Module • Usage & Rider Analytics Module • Integration & Testing • Presentation
	16/12-22/12	1 – MVP Stabilization 2 – Review & Second-Semester Planning

CALENDAR

Milestone	Date	Tasks
	23/12-09/02	Christmas break & Exam period — code refactoring
	10/02-23/02	1 - Extensible API Framework & ML Module Integration (Demand Forecasting / Anomaly Detection)
	24/02-09/03	2 - Operator Backoffice Dashboard UI (Telemetry & Service KPIs)
	10/03-23/03	3 - BI Reporting Module Integration
	24/03-02/04	Easter Holidays — code refactoring
	05/04-18/04	4 - Advanced Analytics & ML Model Refinement

CALENDAR

Milestone	Date	Tasks
	19/04-25/04	5 - Integration, Testing & Bug Fixing
	26/04-30/04	Academic Week — development suspension
	03/05-16/05	6 - User Testing 7 - Apply Improvements
	17/05-30/05	8 - Data Collection & Evaluation 9 - Stabilize Final Product
	31/05-03/06	10 - Create Documentation & Technical Report (Guilherme) 11 - Prepare Final Presentation and Defense

CALENDAR

Milestone	Date	Tasks
STUDENTS @DETI	04/06	• Demo • Poster • Video • Technical Report

TASKS

To Do 6

Req: Levantamento e especificação de Requisitos Funcionais

ms-2

☒ KIQ-10

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MM

Subtasks 0/2

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Req: Definição Personas e Casos de Uso

ms-2

☒ KIQ-11

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GG

Subtasks 0/2

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Req: Especificação de Requisitos Não-Funcionais

ms-2

☒ KIQ-12

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JT

Subtasks 0/2

>

Arch: Desenho inicial da Arquitetura Event-Driven

ms-2

☒ KIQ-13

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SG

Subtasks 0/3

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Arch: Desenho da Arquitetura e Setup do Kafka/ MQTT

ms-2

☒ KIQ-14

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SG

Subtasks 0/3

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Slides: Preparar apresentação da MS2

ms-2

☒ KIQ-17

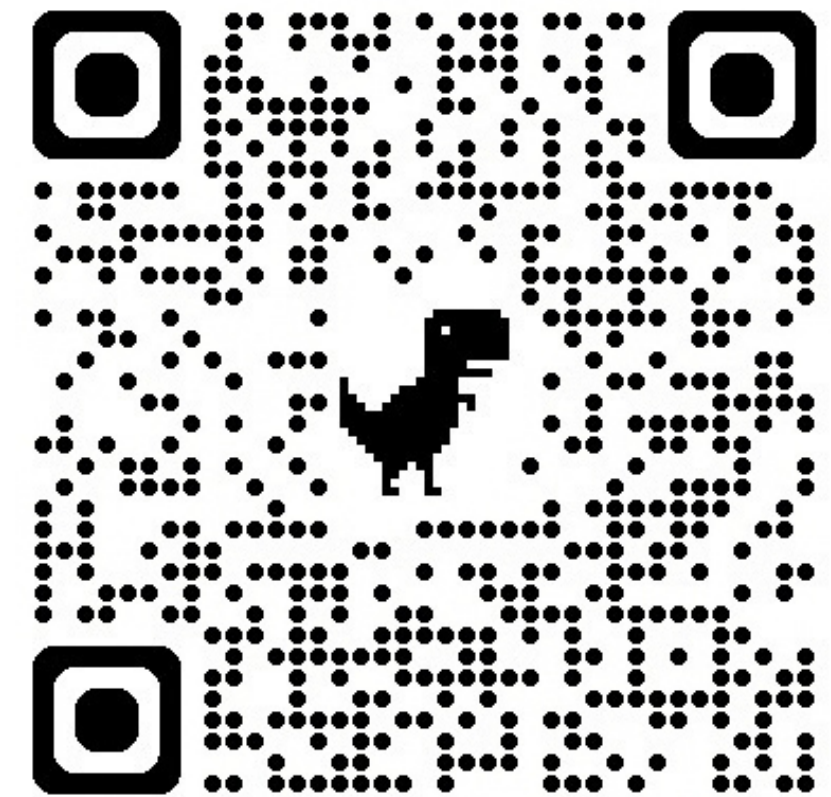
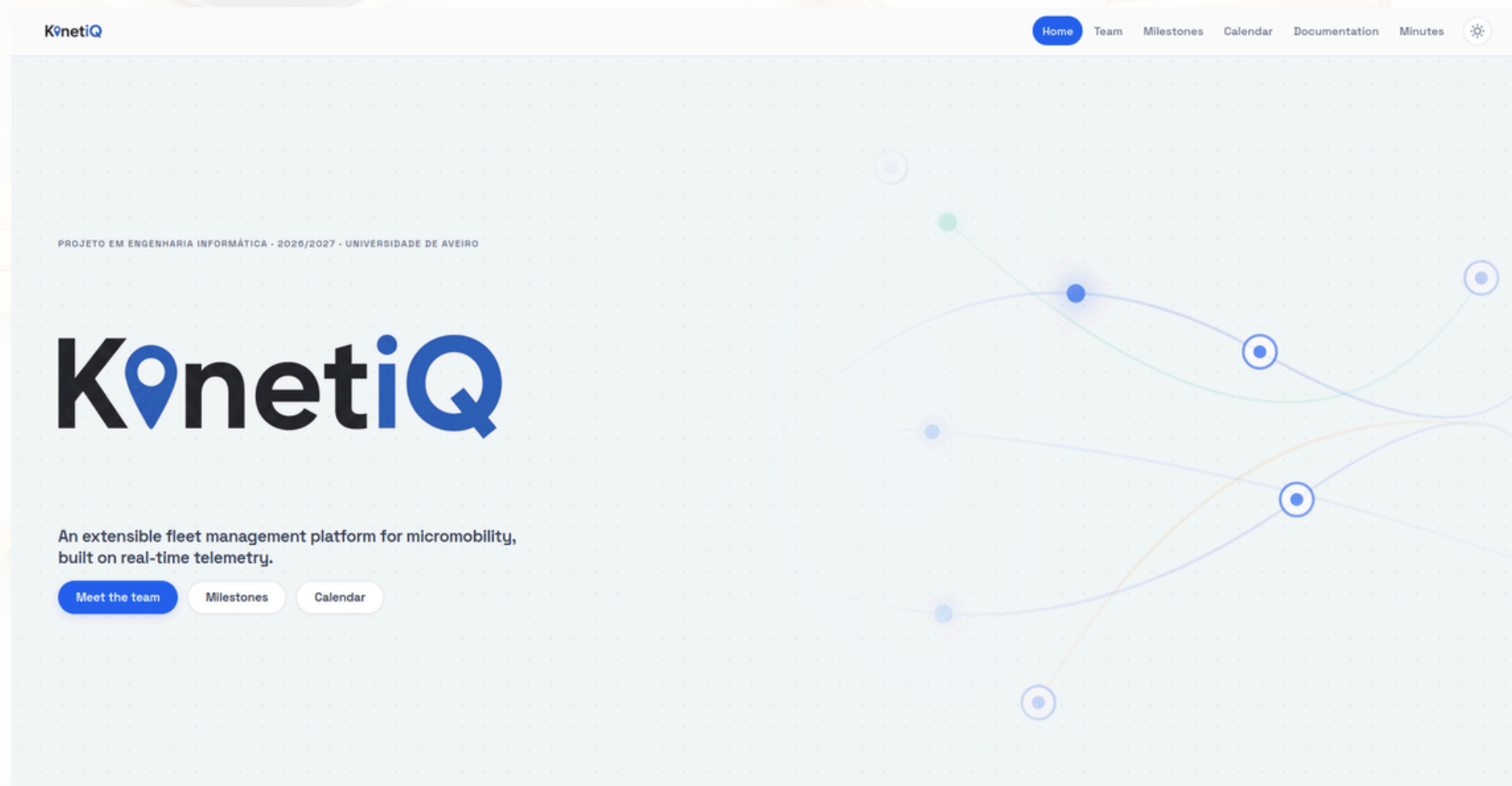
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GG

Subtasks 0/3

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DRAFT VERSION



[HTTPS://KINETIQ-PE1.GITHUB.IO/KINETIQ-SITE/](https://kinetiq-pe1.github.io/kinetiq-site/)



THANK YOU